RealMan Robot rm_ros_interface User Manual V1.3

RealMan Intelligent Technology (Beijing) Co., Ltd.

Revision History

No.	Date	Comment
V1.0	2024-2-18	Draft
V1.1	2024-7-8	Amend(Add teaching messages)
V1.2	2024-12-25	Amend(Add UDP reporting messages)
V1.3	2025-4-3	Amend(Adapt to API2)

目录

rm_ros_interface_Package_Description

The main function of the rm_ros_interface package is to provide necessary message files for the robotic arm to run under the framework of ROS2. In the following text, we will provide a detailed introduction to this package through the following aspects.

- 1.Package use.
- 2.Package architecture description.
- 3.Package topic description.

Through the introduction of the three parts, it can help you-

- 1.Understand the package use.
- 2.Familiar with the file structure and function of the package.
- 3.Familiar with the topic related to the package for easy development and use.

rm_ros_interface_Package_Use

This package does not have any executable commands, but it is used to provide the necessary message files for other packages.

rm_ros_interface_Package_Architecture_Description

Overview_of_Package_Files

```
├── CMakeLists.txt                # compilation rule file
├── include                       # dependency header file folder
│   └── rm_ros_interfaces
├── msg                          # current message file (see below for details)
│   ├── Armcurrentstatus.msg
│   ├── Armoriginalstate.msg
│   ├── Armstate.msg
│   ├── Cartepos.msg
│   ├── Carteposcustom.msg
│   ├── Forcepositionmovejoint.msg
│   ├── Forcepositionmovepose.msg
│   ├── Force_Position_State.msg
│   ├── Getallframe.msg
│   ├── GetArmState_Command.msg
│   ├── Gripperpick.msg
│   ├── Gripperset.msg
│   ├── Handangle.msg
│   ├── Handforce.msg
│   ├── Handposture.msg
│   ├── Handseq.msg
│   ├── Handspeed.msg
│   ├── Handstatus.msg
│   ├── Jointcurrent.msg
│   ├── Jointenflag.msg
│   ├── Jointerrclear.msg
│   ├── Jointerrorcode.msg
│   ├── Jointposeeuler.msg
│   ├── Jointpos.msg
│   ├── Jointposcustom.msg
│   ├── Jointspeed.msg
│   ├── Jointteach.msg
│   ├── Jointtemperature.msg
│   ├── Jointvoltage.msg
│   ├── Liftheight.msg
│   ├── Liftspeed.msg
│   ├── Liftstate.msg
│   ├── Movec.msg
│   ├── Movej.msg
│   ├── Movejp.msg
│   ├── Movel.msg
│   ├── Ortteach.msg
│   ├── Posteach.msg
│   ├── Setforceposition.msg
│   ├── Setrealtimepush.msg
│   ├── Sixforce.msg
```

```
│   └── Stop.msg
├── package.xml                                    # dependency declaration file
└── src
```

rm_ros_interface_message_description

Joint_error_code-Jointerrorcode_msg

```
uint16[] joint_error
uint8 dof
```

msg member uint16[] joint_error Error message for each joint. uint8 dof Degree of freedom message of the robotic arm.

Clearing_the_joint's_error_code-Jointerrclear_msg

```
uint8 joint_num
```

msg member joint_num the corresponding joint number, from the base to the robotic arm gripper, the number is 1-6 or 1-7.

All_coordinate_system_names-Getallframe_msg

```
string[10] frame_name
```

msg member frame_name The array of work coordinate system names returned

Joine_motion-Movej_msg

```
float32[] joint
uint8 speed
bool block
uint8 trajectory_connect
uint8 dof
```

msg member joint Joint angle, float type, unit-radians. speed Speed percentage ratio coefficient, 0-100. trajectory_connect Is the trajectory plan now. 1.wait 0.plan now. block whether it is a blocking mode, bool type, true-blocking, false-non-blocking. dof Degree of freedom message of the robotic arm.

Linear_motion-Movel_msg

```
geometry_msgs/Pose pose
uint8 speed
uint8 trajectory_connect
bool block
```

msg member pose Robotic arm pose-geometry_msgs/Pose type, x, y, z coordinates (float type, unit-m) + quaternion (float type). speed Speed percentage ratio coefficient, 0-100. trajectory_connect Is the trajectory plan now. 1.wait 0.plan now. block whether it is a blocking mode, bool type, true-blocking, false-non-blocking.

Circular_motion-Movec_msg

```
geometry_msgs/Pose pose_mid
geometry_msgs/Pose pose_end
uint8 speed
uint8 trajectory_connect
bool block
uint8 loop
```

msg member pose_mid Middle pose: geometry_msgs/Pose type, x, y, z coordinates (float type, unit: m) + quaternion. pose_end Target pose: geometry_msgs/Pose type, x, y, z coordinates (float type, unit: m) + quaternion. speed Speed percentage ratio coefficient, 0-100. trajectory_connect Is the trajectory plan now. 1.wait 0.plan now. block whether it is a blocking mode, bool type, true-blocking, false-non-blocking. loop Number of cycles

Joint_space_planning_to_target_pose-Movejp_msg

```
geometry_msgs/Pose pose
uint8 speed
uint8 trajectory_connect
bool block
```

msg member pose Target pose: geometry_msgs/Pose type, x, y, z coordinates (float type, unit: m) + quaternion. speed Speed percentage ratio coefficient, 0-100. trajectory_connect Is the trajectory plan now. 1.wait 0.plan now. block whether it is a blocking mode, bool type, true-blocking, false-non-blocking.

Joint_teaching-Jointteach_msg

```
uint8 num
uint8 direction
uint8 speed
```

msg member num joint num，1~7. direction teach direction，0-negative direction，1-positive direction. speed speed:speed percentage ratio coefficient, 0-100.

Position_teaching-Posteach_msg

```
uint8 type
uint8 direction
uint8 speed
```

msg member type Teaching demonstration type: input0:X-axis direction、1:Y-axis direction、2:Z-axis direction. direction teach direction，0-negative direction，1-positive direction. speed speed:speed percentage ratio coefficient, 0-100.

Attitude_teaching-Ortteach_msg

```
uint8 type
uint8 direction
uint8 speed
```

msg member type Teaching demonstration type: input0:RX-axis direction、1:RY-axis direction、2:RZ-axis direction direction teach direction，0-negative direction，1-positive direction. speed speed:speed percentage ratio coefficient, 0-100.

Joint_transmission-Jointpos_msg

```
float32[] joint
bool follow
float32 expand
uint8 dof
```

msg member joint Joint angle, float type, unit: radians. follow Follow state, bool type, true: high follow, false: low follow, default high follow if not set. expand Expand joint, float type, unit: radians. dof Degree of freedom message of the robotic arm.

Pose_transmission-Cartepos_msg

```
geometry_msgs/Pose pose
bool follow
```

msg member pose Robotic arm poses geometry_msgs/Pose type, x, y, z coordinates (float type, unit: m) + quaternion. follow Follow state, bool type, true: high follow, false: low follow, default high follow if not set.

Current_robotic_arm_state_Angle_and_Euler_angle-Armoriginalstate_msg

```
float32[] joint
float32[6] pose
uint16 arm_err
uint16 sys_err
uint8 dof
```

msg member joint Joint angle, float type, unit: °. pose Current pose of the robotic arm, float type, x, y, z coordinates, unit: m, x, y, z Euler angle, unit: degree. arm_err Robotic arm running error code, unsigned int type. arm_err Controller error code, unsigned int type. dof Degree of freedom message of the robotic arm.

Current_arm_state_radians_and_quaternion-Armstate_msg

```
float32[] joint
geometry_msgs/Pose pose
uint16 arm_err
uint16 sys_err
uint8 dof
```

msg member joint Joint angle, float type, unit: radians. pose Current pose of the robotic arm, float type, x, y, z coordinates, unit: m, x, y, z, w quaternion. arm_err Robotic arm running error code, unsigned int type. arm_err Controller error code, unsigned int type. dof Degree of freedom message of the robotic arm.

Gripper_pick-Gripperpick_msg

```
uint16 speed
uint16 force
bool block
uint16 timeout
```

msg member speed Gripper pick speed, unsigned int type, range: 1-1000. force Gripper pick torque threshold, unsigned int type, range: 50-1000. block whether it is a blocking mode, bool type, true-blocking, false-non-blocking. timeout Set the timeout for the return, and the blocking mode will take effect (in seconds).

Gripper_pick_gripper_pick-on-Gripperpick_msg

```
uint16 speed
uint16 force
bool block
uint16 timeout
```

msg member speed Gripper pick speed, unsigned int type, range: 1-1000. force Gripper picks torque threshold, unsigned int type, range: 50-1000. block whether it is a blocking mode, bool type, true-blocking, false-non-blocking. timeout Set the timeout for the return, and the blocking mode will take effect (in seconds).

Gripper_reaching_the_given_position-Gripperset_msg

```
uint16 position
bool block
uint16 timeout
```

msg member position Gripper target position, unsigned int type, range: 1-1000, representing the degree of opening of the gripper: 0-70 mm. block whether it is a blocking mode, bool type, true-blocking, false-non-blocking. timeout Set the timeout for the return, and the blocking mode will take effect (in seconds).

Force-position_mixing_control-Setforceposition_msg

```
uint8 sensor
uint8 mode
uint8 direction
int16 n
```

msg member sensor Sensor; 0 - One-axis force; 1 - Six-axis force. mode Mode: 0 - Base coordinate system force control; 1 - Tool coordinate system force control. Direction Force control direction; 0 - Along the X-axis; 1 - Along the Y-axis; 2 - Along the Z-axis; 3 - Along the RX posture direction; 4 - Along the RY posture direction; 5 - Along the RZ posture direction. n Force value, unit: 0.1 N.

Six-axis_force_data-Sixforce_msg

```
float32 force_fx
float32 force_fy
float32 force_fz
float32 force_mx
float32 force_my
float32 force_mz
```

msg member force_fx the force along the x-axis direction. force_fy the force along the y-axis direction. force_fz the force along the z-axis direction. force_mx the force when rotating along the x-axis direction. force_my the force when rotating along the y-axis direction. force_mz the force when rotating along the z-axis direction.

Setting_the_dexterous_hand_posture-Handposture_msg

```
uint16 posture_num
bool block
uint16 timeout
```

msg member posture_num The serial number of the posture pre-saved in the dexterous hand, ranges from 1 to 40. block whether it is a blocking mode, bool type, true-blocking, false-non-blocking. timeout The timeout setting for blocking mode, unit: seconds.

Setting_the_dexterous_hand_action_sequence-Handseq_msg

```
uint16 seq_num
bool block
unint16 timeout
```

msg member seq_num The serial number of the sequence pre-saved in the dexterous hand, ranging from 1 to 40. block whether it is a blocking mode, bool type, true-blocking, false-non-blocking. timeout The timeout setting for blocking mode, unit: seconds.

Setting_the_angles_of_various_degrees_of_freedom_for_the_dexterous_hand-Handangle_msg

```
int16[6] hand_angle
bool block
```

msg member hand_angle Hand angle array, range: 0-1000. And -1 represents that no operation is performed on this degree of freedom and the current state remains. block whether it is a blocking mode, bool type, true-blocking, false-non-blocking.

Setting_the_dexterous_hand_action_sequence-Handspeed_msg

```
uint16 hand_speed
```

msg member hand_speed Hand speed, range: 1-1000.

Setting_the_force_threshold_for_the_dexterous_hand-Handforce_msg

```
uint16 hand_force
```

msg member hand_force Hand force, range: 1-1000.

Transmissive_force-position_mixing_control_compensation-angle-Forcepositionmovejoint_msg

```
float32[] joint
uint8 sensor
uint8 mode
int16 dir
float32 force
bool follow
uint8 dof
```

msg member joint Angle force-position mixing transmission, unit: radians. sensor Type of sensor used, 0 - One-axis force, 1 - Six-axis force. mode Mode, 0 - Along the work coordinate system, 1 - Along the tool end coordinate system. dir Force control direction, 0 to 5 represent X/Y/Z/Rx/Ry/Rz respectively, and the default direction for one-axis force type is the Z direction. force Force value, accuracy: 0.1 N or 0.1 Nm. block whether it is a blocking mode, bool type, true-blocking, false-non-blocking. dof Degree of freedom message of the robotic arm.

Transmissive_force-position_mixing_control_compensation-pose-Forcepositionmovejoint_msg

```
geometry_msgs/Pose pose
uint8 sensor
uint8 mode
int16 dir
float32 force
bool follow
```

msg member pose Robotic arm pose message, x, y, z position message + quaternion posture message. sensor Type of sensor used, 0 - One-axis force, 1 - Six-axis force. mode Mode, 0 - Along the work coordinate system, 1 - Along the tool end coordinate system. dir Force control direction, 0 to 5 represent X/Y/Z/Rx/Ry/Rz respectively, and the default direction for one-axis force type is the Z direction. force Force value, accuracy: 0.1 N or 0.1 Nm. block whether it is a blocking mode, bool type, true-blocking, false-non-blocking.

Speed_open_loop_control-lifting_mechanism-Liftspeed_msg

```
int16 speed
bool block
```

msg member speed Speed percentage, -100-100. Speed 0: the lifting mechanism moves upward; Speed = 0: the lifting mechanism stops. block whether it is a blocking mode, bool type, true-blocking, false-non-blocking.

Position_closed-loop_control-lifting_mechanism-Lift_height_msg

```
uint16 height
uint16 speed
bool block
```

msg member height Target height, unit: mm, accuracy: 0-2600. speed Speed percentage, 1-100. block whether it is a blocking mode, bool type, true-blocking, false-non-blocking.

Getting_the_state_of_the_lifting_mechanism-Liftstate_msg

```
int16 height
int16 current
uint16 err_flag
```

msg member height Current lifting mechanism height, unit: mm, accuracy: 1mm, range: 0-2300. current Lifting drive error code, error code type refers to joint error code.

Getting_or_setting_UDP_active_reporting_configuration-Setrealtimepush_msg

```
uint16 cycle
uint16 port
uint16 force_coordinate
string ip
bool aloha_state_enable
bool arm_current_status_enable
bool expand_state_enable
bool hand_enable
bool joint_speed_enable
bool lift_state_enable
bool plus_base_enable
bool plus_state_enable
```

msg member cycle Set the broadcast cycle, which is a multiple of 5ms. port Set the port number for broadcasting. force_coordinate Coordinate system for external force data of the system, where 0 is the sensor coordinate system, 1 is the current work coordinate system, and 2 is the current tool coordinate system. ip Customized reporting target IP address. aloha_state_enable aloha master arm state enable. arm_current_status_enable arm_current_status enable. expand_state_enable Expansion joint-related data enable. hand_enable Dexterous hand state enable. joint_speed_enable Current joint speed, with an accuracy of 0.02 rpm. lift_state_enable Lifting joint data enable. plus_base_enable Basic information of the end-effector device enable. plus_state_enable Real-time information of the end-effector device enable.

UDP_manipulator_status_report-Armcurrentstatus_msg

```
uint16 arm_current_status
```

msg member arm_current_status Mechanical arm status, 0-RM_IDLE_E // Enabled but idle state 1-RM_MOVE_L_E // move L state in motion 2-RM_MOVE_J_E // move J in motion 3-RM_MOVE_C_E // move C state in motion 4-RM_MOVE_S_E // move S state in motion 5-RM_MOVE_THROUGH_JOINT_E // Angle transparent state 6-RM_MOVE_THROUGH_POSE_E // Pose transparent state 7-RM _ move _ through _ force _ pose _ e//Force control transparent transmission state 8-RM_MOVE_THROUGH_CURRENT_E // Current loop transparent state 9-RM_STOP_E // Emergency stop status 10-RM_SLOW_STOP_E // slow stop status 11-RM_PAUSE_E // Pause status 12-RM_CURRENT_DRAG_E // Current loop drag status 13-RM_SENSOR_DRAG_E // Six-axis force drag state 14-RM_TECH_DEMONSTRATION_E // Teaching Status

UDP_joint_current_report-Jointcurrent_msg

```
float32[] joint_current
```

msg member joint_current Current joint current with an accuracy of 0.001mA.

UDP_joint_enabling_status_report-Jointenflag_msg

```
bool[] joint_en_flag
```

msg member joint_en_flag Current joint enabling state, 1 is up enabling and 0 is down enabling.

UDP_manipulator_Eulers_angular_pose_is_reported_to-Jointposeeuler_msg

```
float32[3] euler
float32[3] position
```

msg member euler Euler angle of current waypoint attitude, with an accuracy of 0.001rad. position The current waypoint position has an accuracy of 0.000001M.

UDP_joint_speed_report_Jointspeed_msg

```
float32[] joint_speed
```

msg member joint_speed Current joint speed, accuracy 0.02RPM.

UDP_joint_temperature_report_Jointtemperature_msg

```
float32[] joint_temperature
```

msg member joint_temperature Current joint temperature, with an accuracy of 0.001℃.

UDP_joint_voltage_report_Jointvoltage_msg

```
float32[] joint_voltage
```

msg member joint_voltage Current joint voltage, with an accuracy of 0.001V V.

System_error_code_Rmerr_msg

```
uint8 err_len
int32[] err
```

msg 成员 err_len uint8。 err int32。

Basic_information_of_the_end_effector_device_Rmplusbase_msg

```
string manu              # Device manufacturer;
int8 type                # Device type, 1 - Two-finger gripper, 2 - Five-finger
dexterous hand, 3 - Three-finger gripper;
string hv                # Hardware version;
string sv                # Software version;
string bv                # Bootloader version;
int32 id                 # Device ID;
int8 dof                 # Degrees of freedom;
int8 check               # Self-check switch;
int8 bee                 # Beeper switch;
bool force               # Force control support;
bool touch               # Tactile support;
int8 touch_num           # Number of tactile sensors;
int8 touch_sw            # Tactile switch;
int8 hand                # Hand orientation, 1 - Left hand, 2 - Right hand;
int32[12] pos_up         # Position upper limit
```

```
int32[12] pos_low       # Position lower limit
int32[12] angle_up      # Angle upper limit 0.01°
int32[12] angle_low     # Angle lower limit 0.01°
int32[12] speed_up      # Speed upper limit
int32[12] speed_low     # Speed lower limit
int32[12] force_up      # Force upper limit 0.001N
int32[12] force_low     # Force lower limit 0.001N
```

Real_time_information_of_the_end_effector_deviceRmplusstate_msg

```
int32 sys_state                  #System status
int32[12] dof_state              #Current status of each degree of freedom (DoF)
int32[12] dof_err                #Error information of each DoF
int32[12] pos                    #Current position of each DoF
int32[12] speed                  #Current Speed of Each Degree of each DoF
int32[12] angle                  #Current Angle of Each Degree of each DoF
int32[12] current                #Current Current of Each Degree of Freedom
int32[18] normal_force           #Normal Force of Tactile Three-Dimensional Force of
Each Degree of Freedom
int32[18] tangential_force       #Tangential Force of Tactile Three-Dimensional Force
of Each Degree of Freedom
int32[18] tangential_force_dir   #Direction of Tangential Force of Tactile Three-
Dimensional Force of Each Degree of Freedom
uint32[12] tsa                   #Tactile Self-Approach of Each Degree of Freedom
uint32[12] tma                   #Tactile Mutual Approach of Each Degree of Freedom
int32[18] touch_data             #Raw Data of Tactile Sensors
int32[12] force                  #Torque of Each Degree of Freedom
```

Customize_high_following_mode_joint_transmission_Jointposcustom_msg

```
float32[] joint
bool follow
float32 expand
uint8 dof
uint8 trajectory_mode
uint8 radio
```

msg member joint Joint angle, float type, unit: radians. follow Follow state, bool type, true: high follow, false: low follow, default high follow if not set. expand Expand joint, float type, unit: radians. dof Degree of freedom message of the robotic arm. trajectory_mode When the high following mode is set, multiple modes are supported, including 0- complete transparent transmission mode, 1- curve fitting mode and 2- filtering mode. radio Set the smoothing coefficient in curve fitting mode (range 0-100) or the filter parameter in filtering mode (range 0-1000). The higher the value, the better the smoothing effect.

Customize_high_following_mode_pose_transmission_Carteposcustom_msg

```
geometry_msgs/Pose pose
bool follow
uint8 trajectory_mode
uint8 radio
```

msg member pose Robotic arm poses geometry_msgs/Pose type, x, y, z coordinates (float type, unit: m) + quaternion. follow Follow state, bool type, true: high follow, false: low follow, default high follow if not set. trajectory_mode When the high following mode is set, multiple modes are supported, including 0- complete transparent transmission mode, 1- curve fitting mode and 2- filtering mode. radio Set the smoothing coefficient in curve fitting mode (range 0-100) or the filter parameter in filtering mode (range 0-1000). The higher the value, the better the smoothing effect.

It is mainly for the application of API to achieve some of the robotic arm functions; for a more complete introduction and use, please see the special document "RealMan Robotic Arm ROS2 Topic Detailed Description (https://github.com/kaola-zero/ros2_rm_robot/blob/main/rm_driver/doc/RealMan%20Robotic%20Arm%20rm_driver%20Topic%20Detailed%20Description%20(ROS2).md)".